

$$V=C[1;5], \langle f, g \rangle = \int_1^5 f(x)g(x)dx, L=\{1, x, x^2\}$$

$$\text{bázy } v_1=1, v_2=x, v_3=x^2$$

$$\text{vektor } e_1: \|v_1\|^2 = \int_1^5 1^2 dx = [x]_1^5 = 4 \quad \|v_1\|^2 = 4 \Rightarrow \|v_1\| = 2$$

$$e_1 = \frac{v_1}{\|v_1\|} = \frac{1}{2}$$

$$e_2 \text{ najprv najst-kolij vektor } u_2 = v_2 - \langle v_2, e_1 \rangle e_1$$

$$e_3: u_3 = v_3 - \langle v_3, e_1 \rangle e_1 - \langle v_3, e_2 \rangle e_2$$

$$\langle v_3, e_1 \rangle = \int_1^5 \frac{x^2}{2} dx = \left[\frac{x^3}{6} \right]_1^5 = \frac{62}{3}$$

$$\langle v_3, e_2 \rangle = \int_1^5 x^2 \cdot \frac{\sqrt{3}}{4} (x-3) dx$$

$$= \frac{\sqrt{3}}{4} \int_1^5 x^3 - 3x^2 dx$$

$$= \frac{\sqrt{3}}{4} \left[\frac{x^4}{4} - x^3 \right]_1^5 = \frac{32 \cdot \sqrt{3}}{4}$$

$$\|u_3\|^2 = \int_1^5 \left(x^2 - 6x + \frac{23}{3} \right)^2 dx$$

$$\|u_3\| = \frac{8}{3 \cdot \sqrt{5}}$$

$$e_3 = \frac{x^2 - 6x + \frac{23}{3}}{\frac{8}{3 \cdot \sqrt{5}}}$$

c)

$$\|\sin x - p(x)\|?$$

nikde používat p vektor

$$\|\sin x - p(x)\|^2 = \|\sin x\|^2 - \|p(x)\|^2$$

$$\|\sin x\|^2 = \left[\frac{x}{2} - \frac{\sin 2x}{4} \right]_1^5 = 2 + \frac{\sin 2 - \sin 10}{4} \approx 2,361$$

na

$$\int_1^5 x \cdot \frac{1}{2} dx = \left[\frac{x^2}{4} \right]_1^5 = \frac{25-1}{4} = 6$$

$$\|u_2\|^2 = \int_1^5 (x-3)^2 dx$$

$$= \left[\frac{(x-3)^3}{3} \right]_1^5 = \frac{8}{3} - \frac{-8}{3} = \frac{16}{3} \Rightarrow \|u_2\| = \frac{4}{\sqrt{3}}$$

$$u_2 = x - \frac{6}{2}$$

$$e_2 = \frac{x-3}{\frac{4}{\sqrt{3}}}$$

b) Trigonometrická projekce

$$f(x) = \sin(x)$$

$$P(x) = c_1 e_1 + c_2 e_2 + c_3 e_3 \quad c_i = \langle \sin x, e_i \rangle$$

$$1. I_0 = \int_1^5 \sin x dx = \cos 1 - \cos 5 = 0,823$$

$$2. I_1 = \int_1^5 x \sin x dx = (\sin 5 - 5 \cos 5) - (\sin 1 - \cos 1) = -7,339$$

$$3. I_2 = \int_1^5 x^2 \sin x dx = [x \sin x - (x^2-2) \cos x]_1^5 = -70,320$$

$$c_1 = \frac{I_0}{2} \quad c_2 = \frac{\sqrt{3}}{4} (I_1 - 3I_0) \quad c_3 = \frac{3 \cdot \sqrt{5}}{8} (I_2 - 6I_1 + \frac{23}{3}I_0)$$

$$p(x) \approx P(x) \approx -0,191x^2 + 1,104x - 0,778$$

$\|p(x)\|^2$ spracovať tiež nástrojom online.

číslo

$$\|\sin x - P(x)\| = \sqrt{2,361 - (c_1^2 + c_2^2 + c_3^2)} \approx 0,035$$

čím je výsledok menší, tým viac je funkcia lepšie aproximovaná